International Data Cycle Project - Group 4

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End-to-end energy & room analytics for HES-SO Bellevue Campus — Sierre, Switzerland. Collects, transforms, and visualizes solar production, power consumption, room bookings, and environmental data through a fully automated daily pipeline.

TECHNICAL OVERVIEW

DATA ANALYTICS

Use case 1: Bellevue Building

Our task was to create an analytics solution utilising the data from different domains related to the Bellevue campus building.

Our end goal was to clean and transform the data in order to visualise it in meaningful dashboards created in PowerBI and SAP Analytics Cloud.

Source: https://www.hevs.ch/fr/hes-so-valais-wallis/campus/campus-bellevue-sierre-205007

Tools Selected

For this project we were free to choose most of our tools ourselves, with the exception of having to use PowerBI and SAC for the dashboards.

We chose to use:

GCP Cloud Storage for the data lake

GCP BigQuery for the data warehouse

GCP Cloud Workflows for orchestrating the pipeline

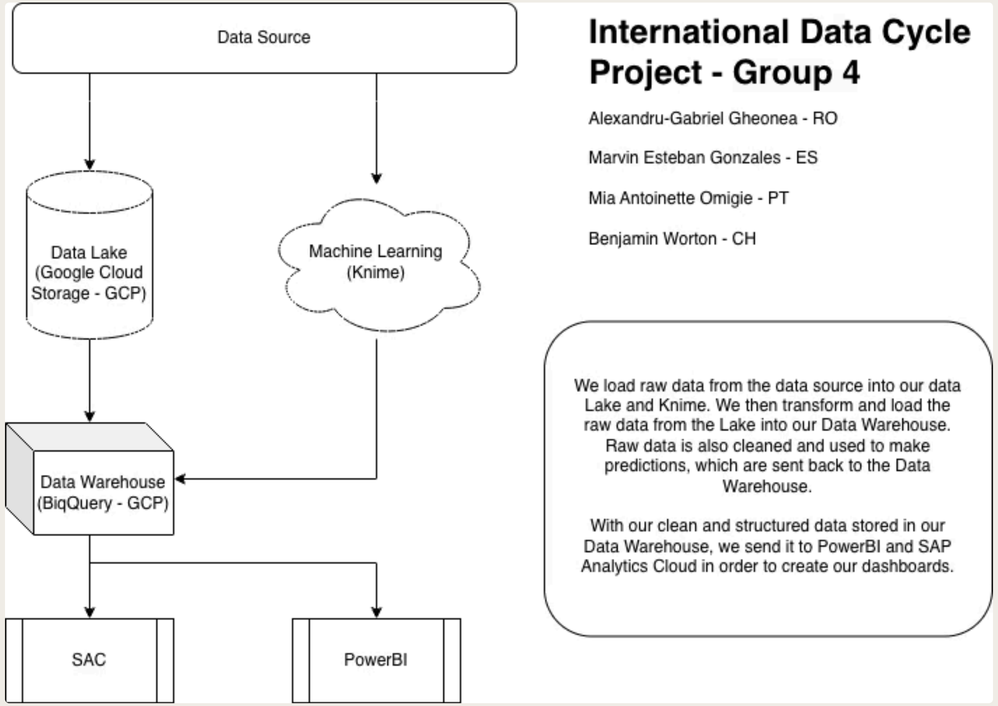
Proton for password management

Knime for machine learning

Pandas for ETL

Jira for project management

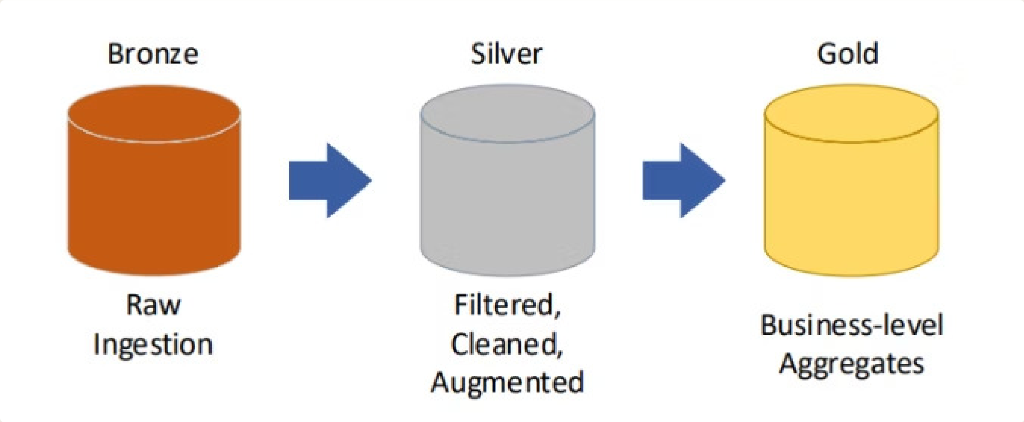
Draw.io for diagrams



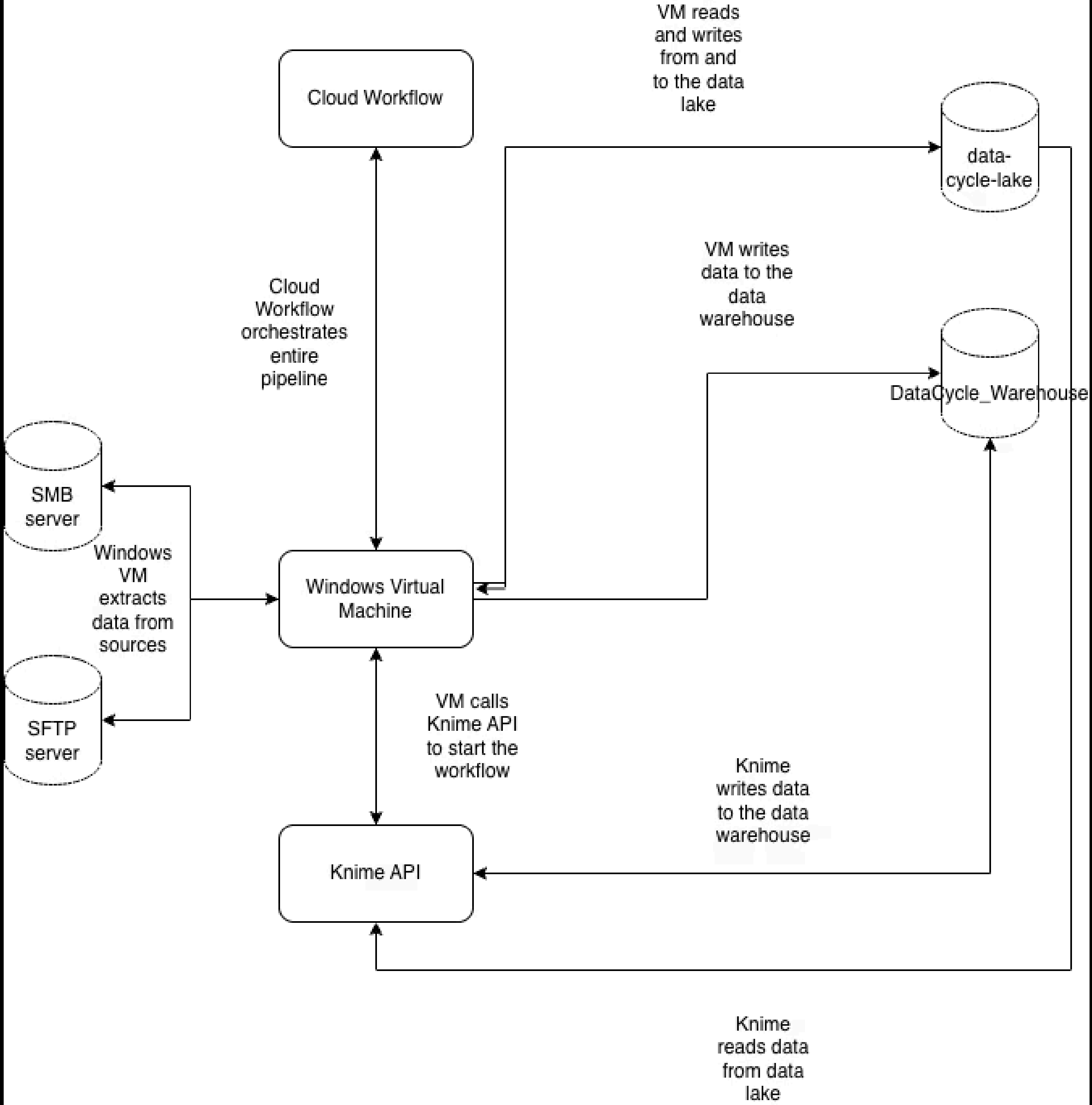
System Architecture

Our solution uses a **medallion architecture** — Bronze → Silver → Gold — orchestrated by GCP Cloud Workflows, with ML predictions via KNIME and visualization through Power BI & SAP Analytics Cloud

All scripts execute daily starting at **9:00 AM**. A Windows VM receives workflow requests via Task Scheduler at 8:50 AM, responding to each step before triggering the next.



Source: Introduction2026.pdf, slide #37, HES-SO Valais / Wallis


VM reads and writes from and to the data lake
Cloud Workflow
data-cycle-lake
VM writes data to the data warehouse
Cloud Workflow orchestrates entire pipeline
DataCycle_Warehouse
SMB server
Windows VM extracts data from sources
Windows Virtual Machine
SFTP server
VM calls Knime API to start the workflow
Knime writes data to the data warehouse
Knime API
Knime reads data from data lake

Bronze Tier — Raw Data Extraction

Scripts extract raw data via **SMB** and **SFTP** from campus servers, storing it in `data-cycle-lake/raw`. Files are organised by month in the data lake.

Solar Production

Inverter logs — historical & current

/raw/solarlogs/production

/raw/solarlogs/productionhistory

Power & Environment

Consumption, temperature & humidity

/raw/bellevueconso/humidity

/raw/bellevueconso/temperature

/raw/bellevueconso/powerconsumption

Room Bookings

CSV & XLS reservation files

/raw/bellevuebooking/csv

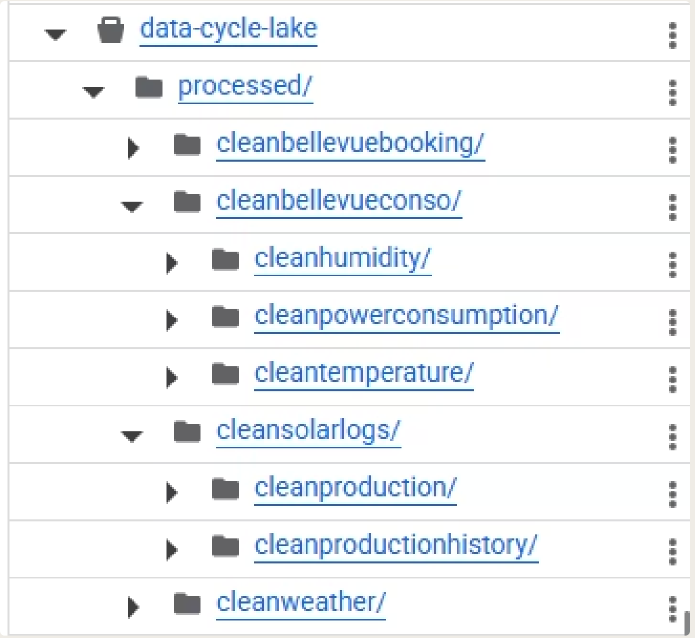
/raw/bellevuebooking/xls

Weather Forecast

SFTP from `/Meteo/`

/raw/weather

Silver Tier — Cleaning & Transformation



Python scripts read Bronze data into **Pandas DataFrames**, clean missing values, deduplicate, and write **Parquet files** with Hive partitioning (`date=yyyy-mm-dd`) to `data-cycle-lake/processed`. This is in order to make future queries faster and to save storage space via compression.

Service account credentials (`SERVICE_ACCOUNT_KEY`) authenticate all GCP operations. Each dataset has a dedicated script — e.g., `PowerConsumptionToSilver.py`.

Gold Tier — BigQuery Data Warehouse

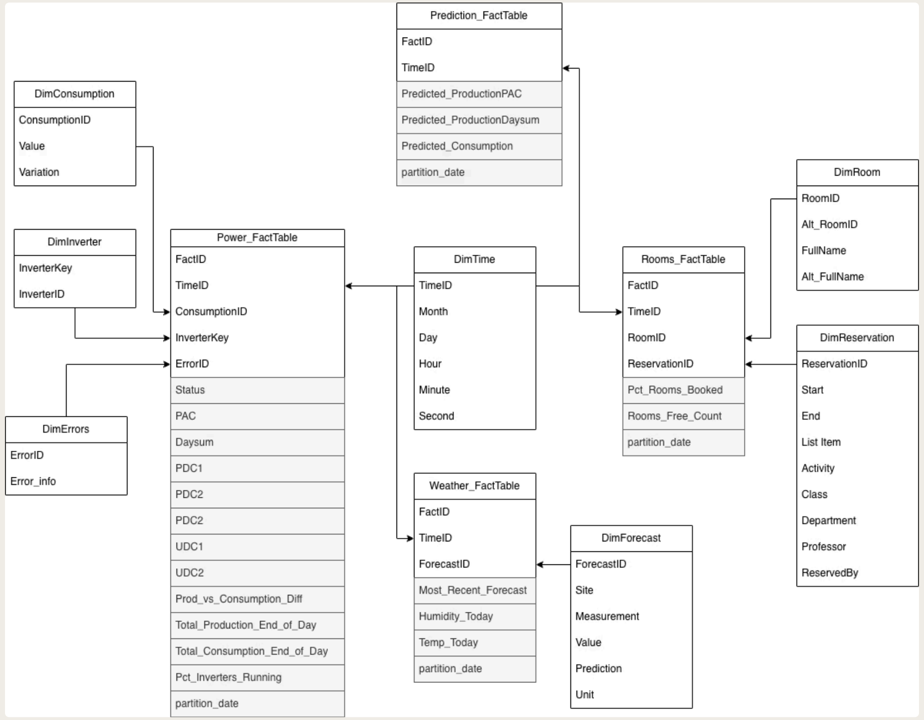
Cleaned Parquet files load into BigQuery (`DataCycle_Warehouse`) using a **galaxy schema** with dimension and fact tables. As BigQuery doesn't support incremental integer IDs, we use 32 character longs UUIDs.

Dimension Tables

- DimTime, DimRoom, DimReservation
- DimInverter, DimForecast, DimErrors

Fact Tables

- Power_FactTable, Prediction_FactTable
- Rooms_FactTable, Weather_FactTable



KNIME — ML Predictions

A deployed KNIME workflow predicts next-day **solar production** and **energy consumption** using Random Forest models, triggered via API after Gold ETL completes.

1 Connect to GCP

Authenticate via service account JSON, read from `data-cycle-lake`

2 Parallel Data Loops

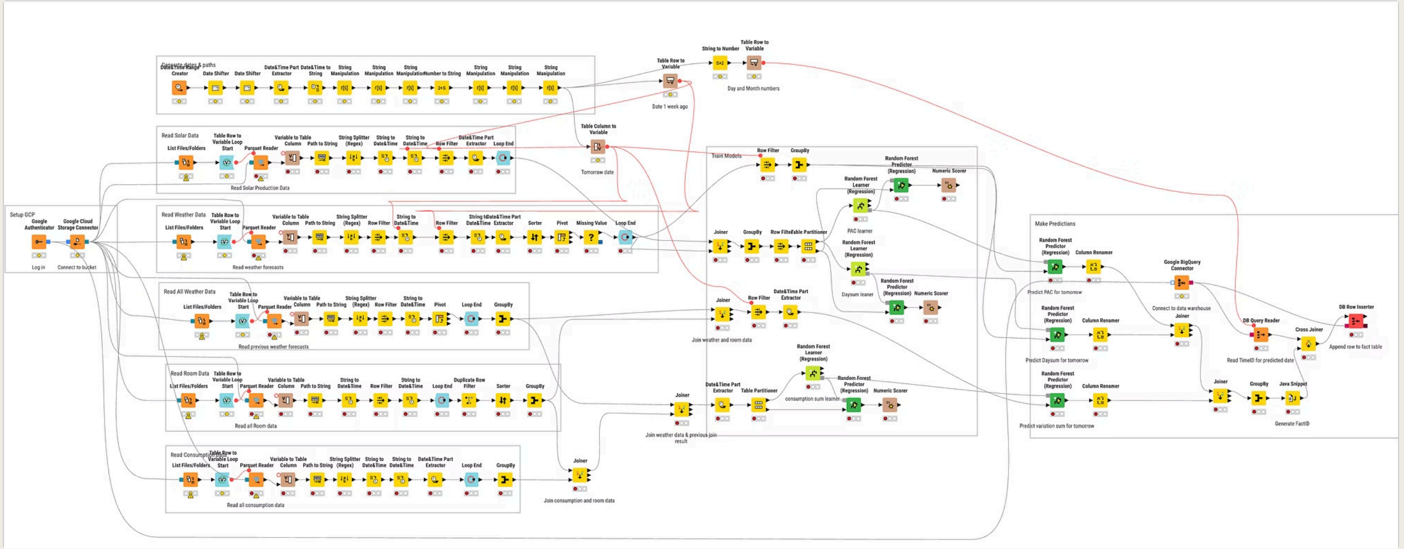
Five loops load bookings, solar, weather, and consumption — weather read twice for different date ranges

3 Train Models

Solar: 7-day rolling window. Consumption: all data from Jan 2023 onward

4 Predict & Store

Append forecasts to `Prediction_FactTable` in BigQuery



Alerts and monitoring

We have alerts set up for when the pipeline doesn't finish executing, and detailed logs for when errors occur. If the Knime workflow fails, it sends an email to a specified address

63	Succeeded	publish_and_wait	send_command	call	Apr 26, 2026, 10:09:10 AM	Apr 26, 2026, 10:09:10 AM
64	Succeeded	publish_and_wait	send_command	call	Apr 26, 2026, 10:09:10 AM	Apr 26, 2026, 10:09:10 AM
65	Succeeded	publish_and_wait	wait_for_vm	call	Apr 26, 2026, 10:09:10 AM	Apr 26, 2026, 10:17:45 AM
66	Succeeded	publish_and_wait	end	return	Apr 26, 2026, 10:17:45 AM	Apr 26, 2026, 10:17:45 AM
67	Failed	main	gold_main	call	Apr 26, 2026, 10:17:45 AM	Apr 26, 2026, 11:17:46 AM
68	Succeeded	publish_and_wait	create_callback	call	Apr 26, 2026, 10:17:45 AM	Apr 26, 2026, 10:17:45 AM
69	Succeeded	publish_and_wait	send_command	call	Apr 26, 2026, 10:17:45 AM	Apr 26, 2026, 10:17:45 AM
70	Succeeded	publish_and_wait	send_command	call	Apr 26, 2026, 10:17:45 AM	Apr 26, 2026, 10:17:45 AM
71	Failed	publish_and_wait	wait_for_vm	call	Apr 26, 2026, 10:17:45 AM	Apr 26, 2026, 11:17:46 AM

Logs of the VM executing the scripts will be saved on the VM, and in the event of errors no duplicates are created and no data is lost.

Power BI Dashboards

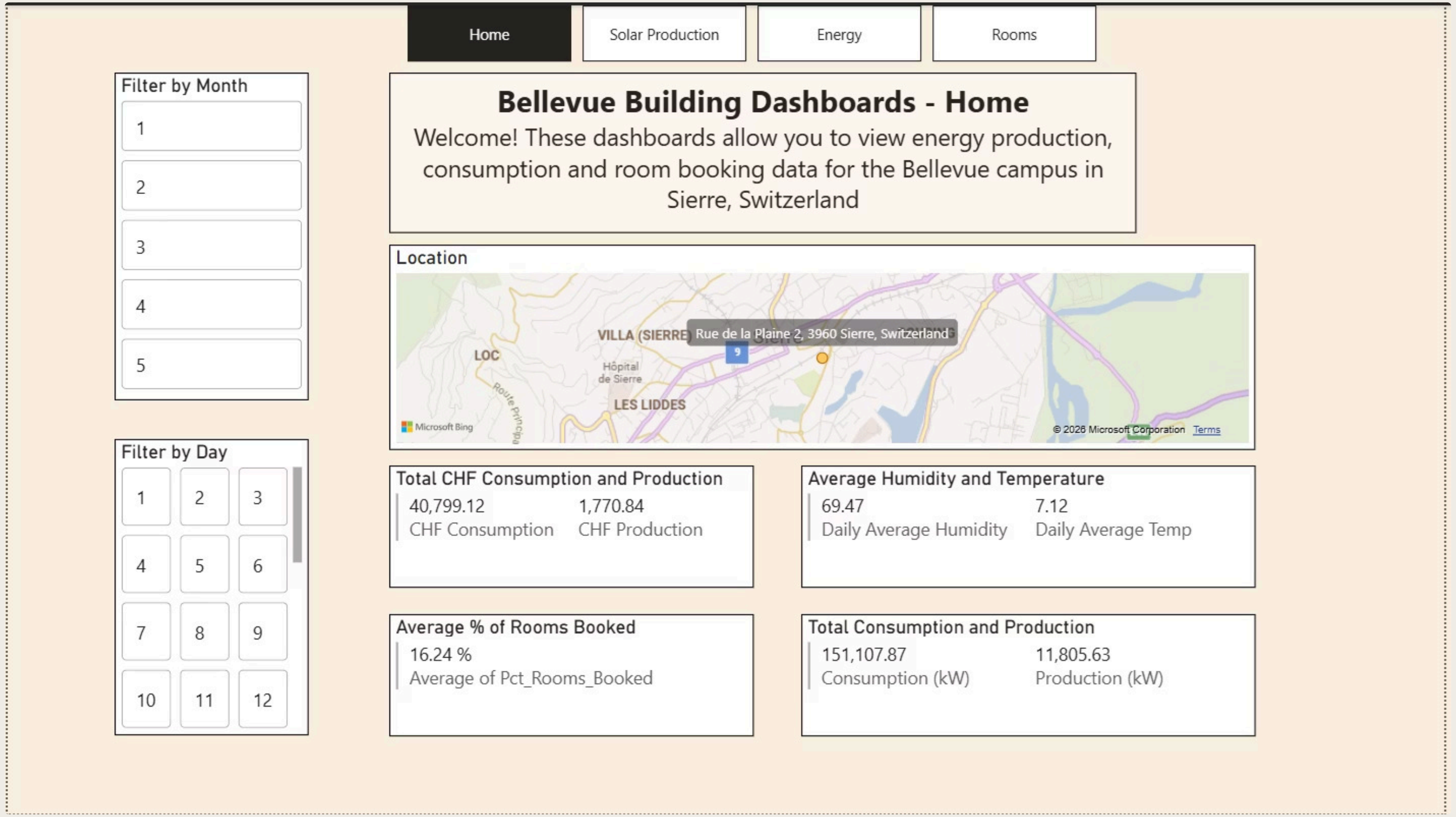
Our solution includes 4 PowerBI dashboards, with 2 roles implemented for row security: **Director** and **Technician**.

Directors have access to all dashboards, whereas Technicians can only view dashboards related to solar production and energy.

SAC Dashboard

Our solution also includes a SAC Dashboard, which is used for visualising data related to inverter performance and errors.

Home Page



The home page acts as the landing page. It features a map element displaying where the building is, as well as 4 cards, which depict:

Total CHF consumption and production

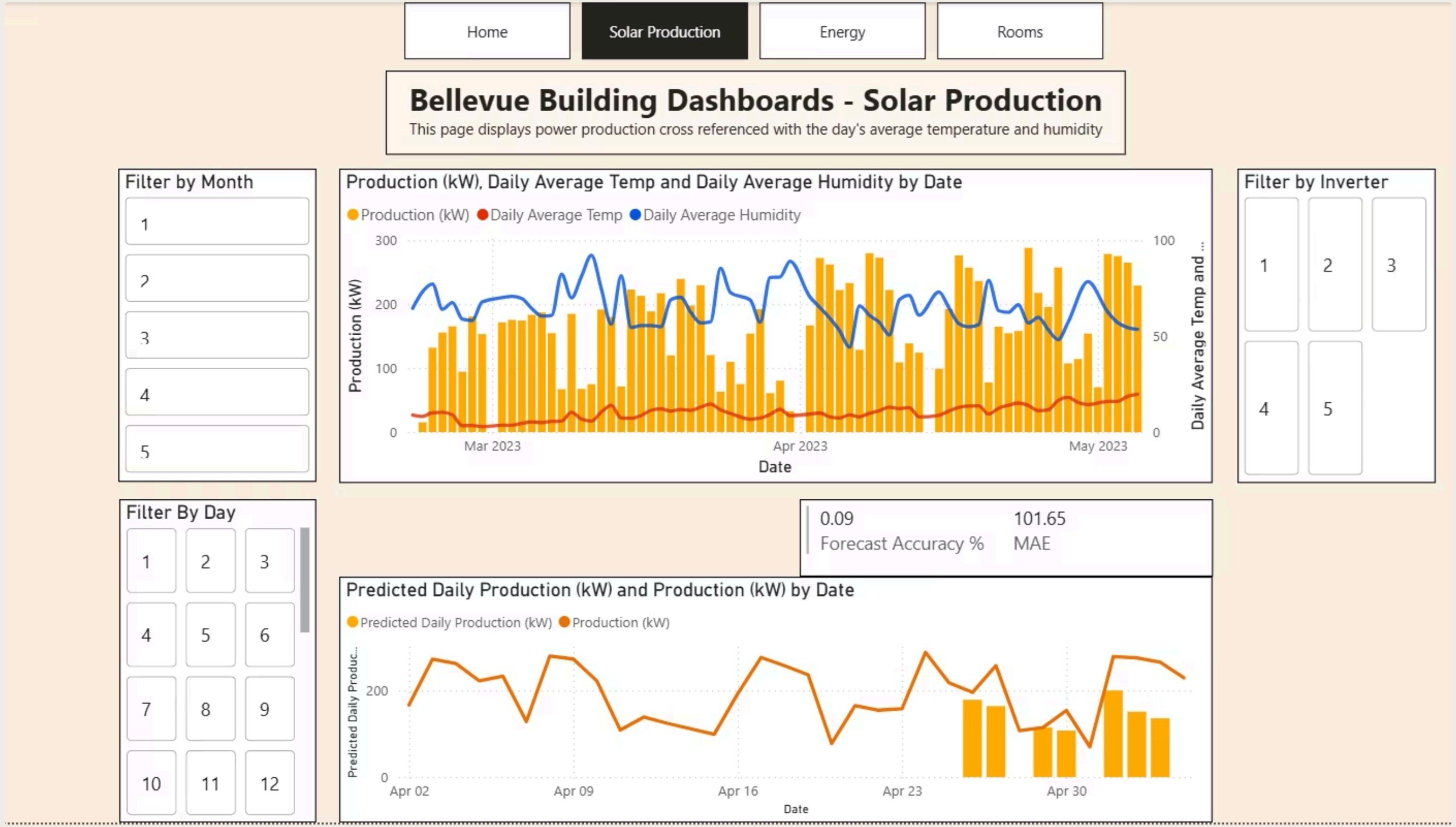
Average humidity and temperature

Average % of rooms booked

Total consumption and production

All cards are filterable by day and month

Solar Production Page



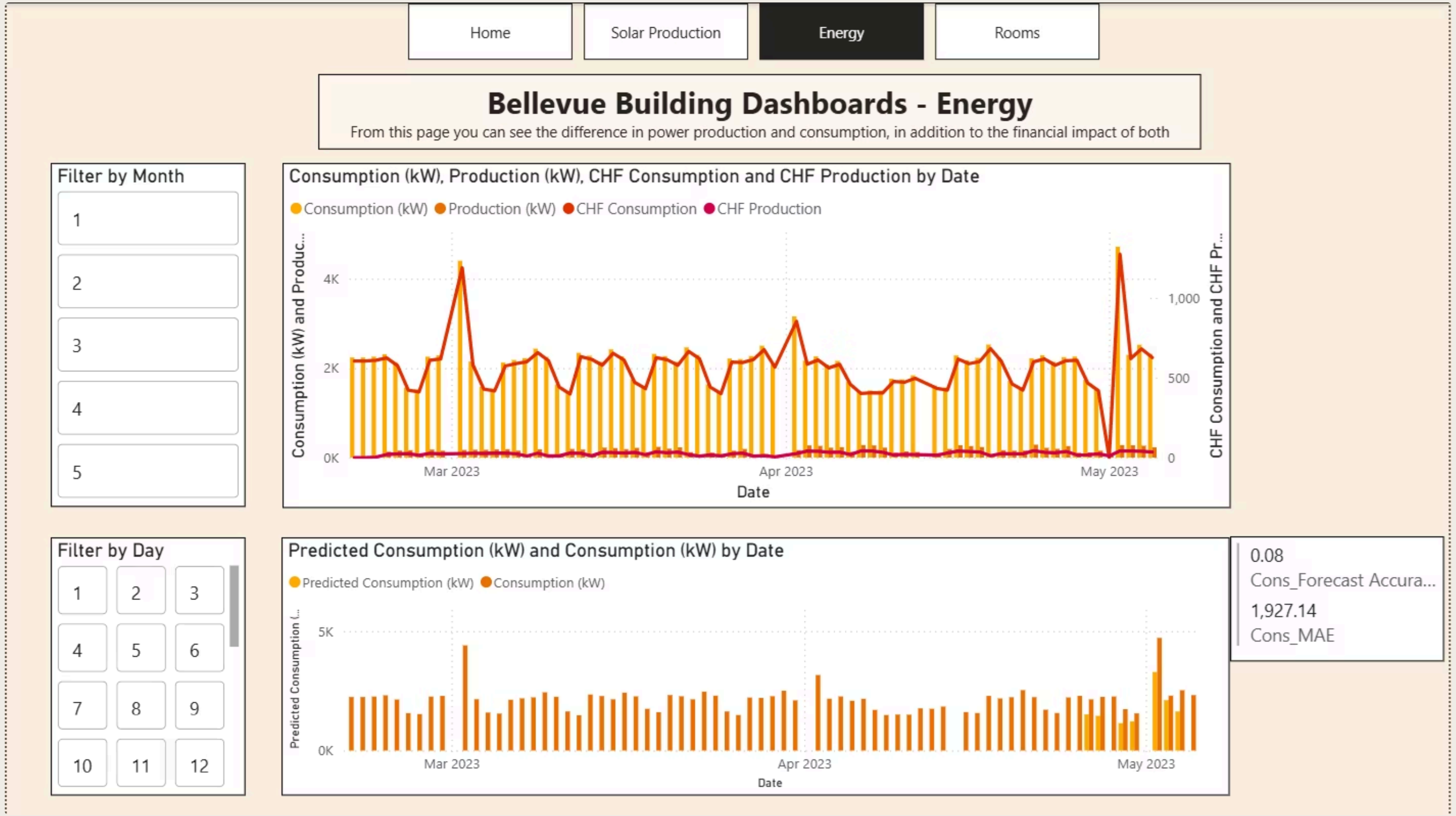
The solar production page has 2 charts.

The first chart displays production over time overlaid with that day's average temperature and humidity.

The second compares the predicted production to the actual production, and includes a card displaying the forecast accuracy and MAE for the selected day

Data is filterable by day, month and inverter

Energy Page



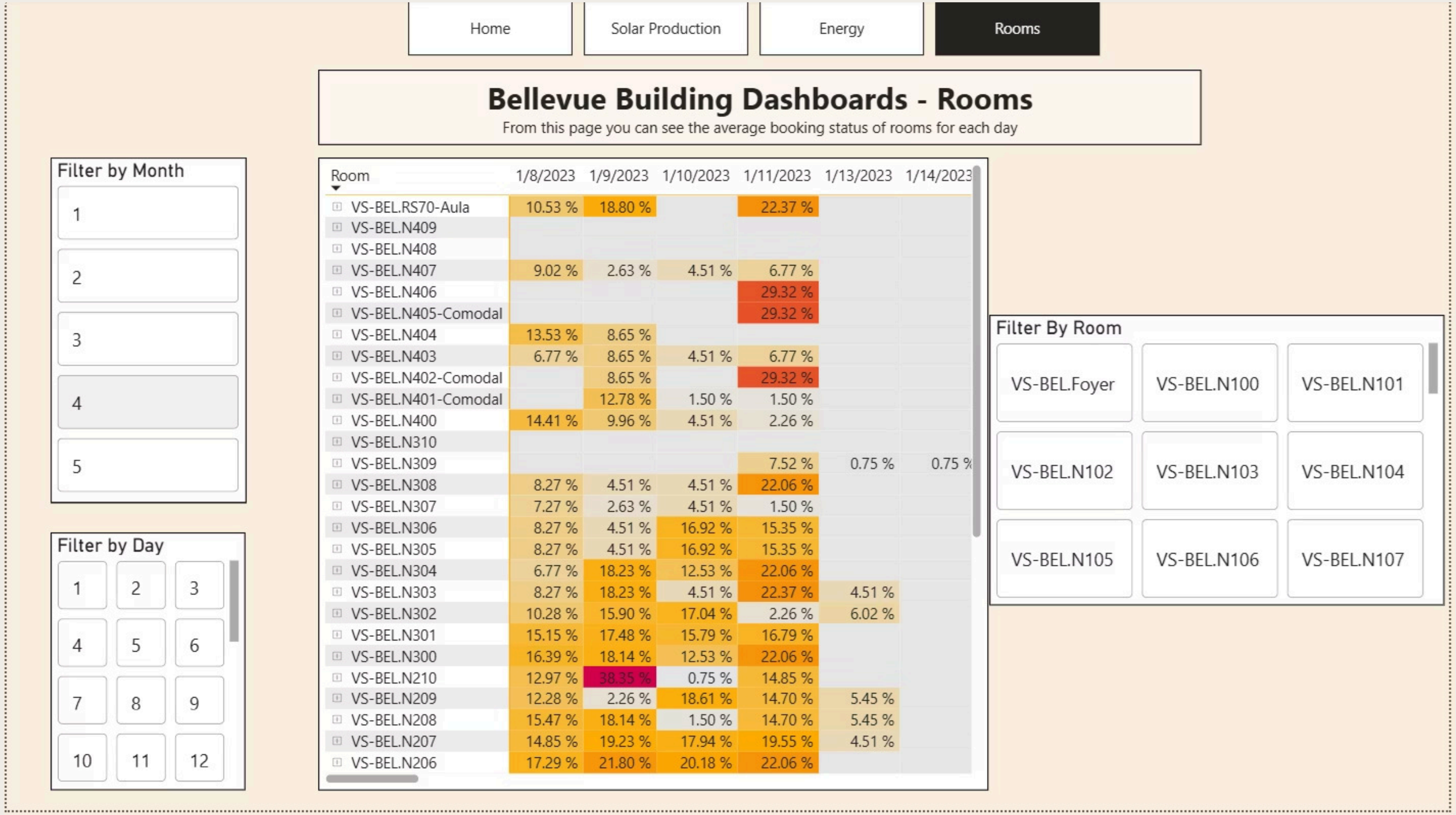
The energy page also includes 2 charts.

The first one compares production to consumption, and also shows the financial impact of both.

The second chart is similar to the other prediction chart, comparing predicted and actual consumption, with a card displaying forecast accuracy and MAE.

Data is filterable by day and month.

Rooms Page

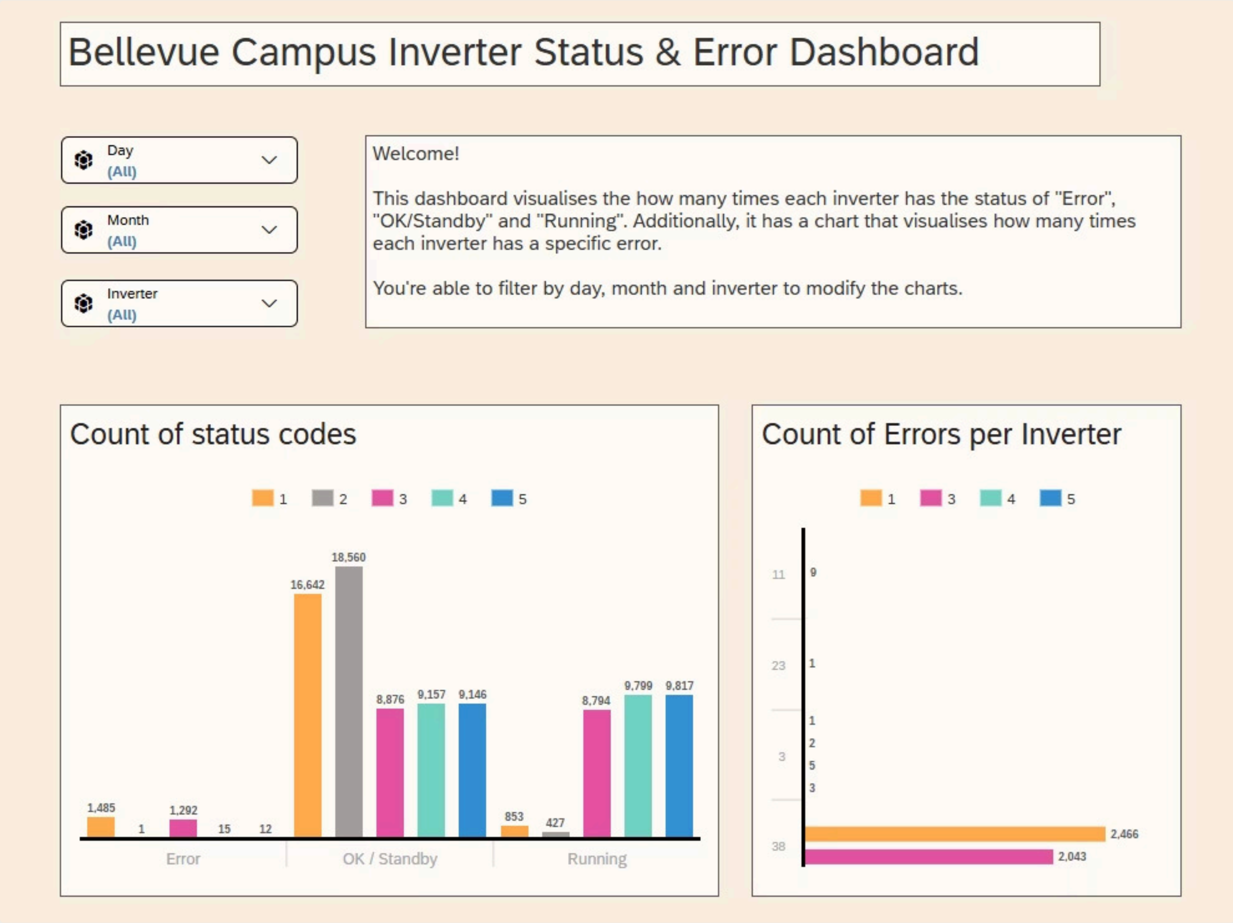


The final PowerBI dashboard is the rooms page.

This page displays each rooms average occupancy during each day. Conditional formatting is applied to the table to create a heatmap, allowing for a quickly interpretable visual indication of how heavily a room is utilised.

Data is filterable by day, month and room.

Inverter Page



The final dashboard is in SAC.
It contains two charts:

1. A chart displaying how many times each inverter has been in an Error, OK / Standby or Running state. This allos us to see which inverters are more prone to errors
2. The second chart displays the count of specific error codes, with 30 being by far the most common

Data is filterable by day, month and inverter

Daily Pipeline Schedule

GCP Cloud Workflows orchestrates the full pipeline automatically each morning. All times are approximate and assume successful execution of each prior step.

1. **08:50 AM**

   Task Scheduler launches `manager.py` on Windows VM which listens for tasks from our workflow in GCP

2. **09:00 AM**

   Cloud Workflow starts — Bronze extraction scripts run

3. **~10:30 AM**

   Solar, consumption, temperature & humidity data available

4. **~10:50 AM**

   Silver cleaning & Gold ETL complete

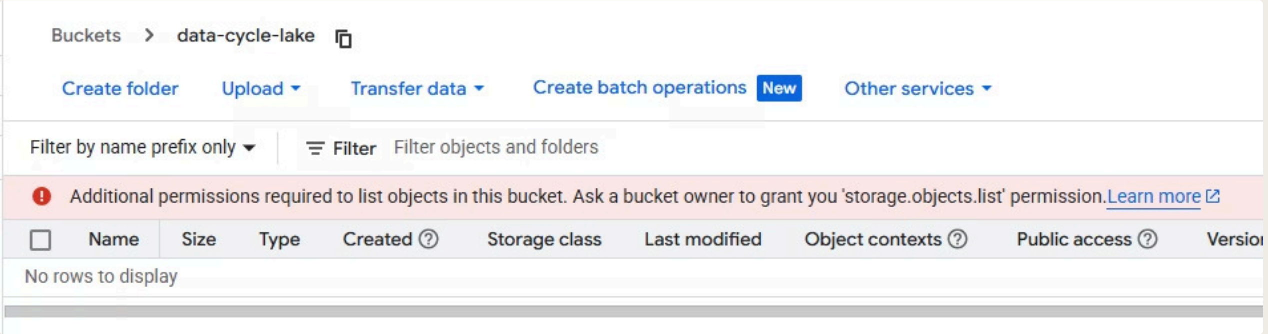
5. **~10:55 AM**

   KNIME predictions appended; Power BI refresh recommended

⚠ Room bookings update **weekly on Monday mornings**.

Security

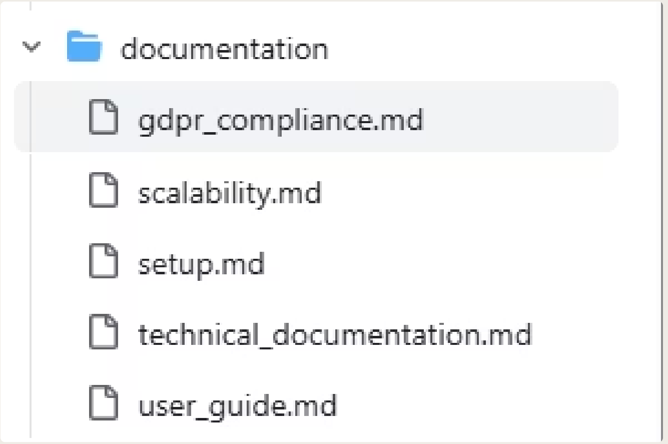
All of our data is encrypted in transit and at rest. Additionally, access to our storage in GCP is limited to only specified users.



PowerBI dashboards have row level security so users only see what they are allowed to see based on their roles, and we manage passwords with Proton password manager.

Personal data is anonymised before it reaches the bronze tier, meaning no personal data is being stored in our solution.

Documentation



Our solution includes extensive documentation to facilitate a hypothetical team's onboarding.

1. Technical documentation that explains the architecture, the workflow, ETL, all scripts in the solution, the machine learning part, the dashboards and security.
2. A scalability assessment of our solution
3. An assessment on if our solutions is GDPR-compliant
4. A setup guide that explains how to set up the solution from scratch
5. A user guide that explains the data and dashboards in-depth

Setup and deployment

Our solution includes a setup guide and automated .bat script to set it up from scratch on a new machine, with the 3 main required resources being:

Windows VM

Install Python, extract the solution ZIP, configure `.env` with HMAC keys and KNIME credentials. Schedule `manager.py` daily via Task Scheduler.

GCP Account

Create a Cloud Storage bucket, BigQuery dataset, Pub/Sub topic, Cloud Workflow, and service account. Store the service account JSON for authentication.

KNIME Hub

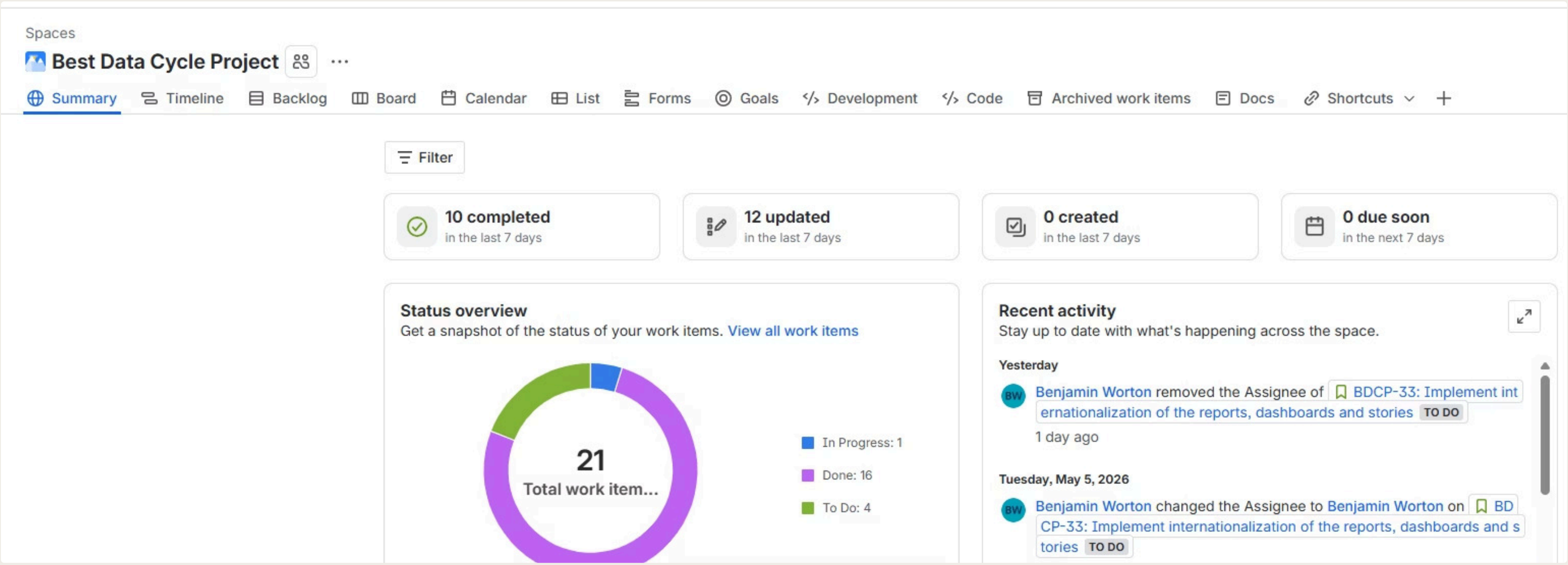
EduHub (academic) or BusinessHub (private). Import `Group4EnergyPredictions.knwf`, deploy as a service, and create a versioned API endpoint.

Project management

Our initial vision for how to manage this project was using Agile development where we used Jira to manage our backlog.

Initially work was done in sprints, but in the latter half of development they were less used as work it did not turn out to be an effective method for delegating work.

Teams was initially used for communication between team members, but the group moved to using a WhatsApp group. Summaries of meetings were provided for members who were not present in meetings, and periodic updates on progress were given.

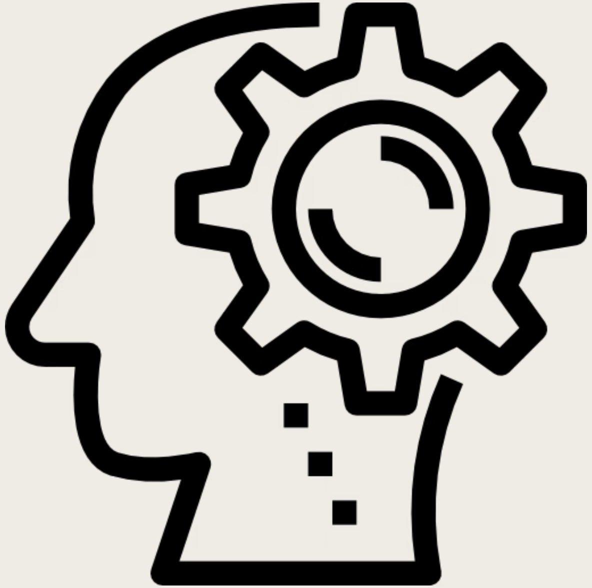


Challenges in this project

The most evident challenges were skill differences, and scheduling issues. Most tools and technologies chosen for this project were new for all members, but different baselines affected each member's ability to participate. This inevitably lead to some members having to compensate for others' workloads.

Additionally, scheduling issues made meetings difficult to organise, thus making work difficult to organise as all communication was asynchronous. Onboarding and explaining topics for members with less experience was subsequently made more difficult due to this.

What have we learned?



- Better understanding of data engineering
- More experience using cloud services
- Do's and dont's for project management
- Better understanding of how to integrate machine learning into data engineering pipelines
- Better understanding of data lake and data warehouse architecture
- More comprehensive understanding of data analytics and visualisation

Thank you for your attention.

Questions?